

Window Functions

- A window function lets you compute a value for each row while "looking at" other rows that are related to it
- Creates a new column when used in the SELECT statement.
- We use it with the `OVER()` statement

Partitions

- is to breakdown things into groups of similarity
- groups data that is similar to one another without collapsing the rows

Syntax:
 window_function (expression)
 OVER (PARTITION BY < grouping columns >
 ORDER BY < sorting columns >) AS < descriptive_name >

the column you want to break down

PARTITION BY - break down each group

ORDER BY - define the sequence

LEAD function

- finds the next row
- looks ahead
- future

LAG function

- finds the previous row
- looks behind
- past

Sales

sale_id	Salesperson	Sale_date	Revenue
1	Mbali	2026-08-01	1000
2	Mbali	2026-08-02	1200
3	Mbali	2026-08-03	900
4	John	2026-08-04	800
5	John	2026-08-05	950

```
SELECT salesperson,
       sale_date,
       revenue,
       LEAD (Revenue)
       OVER (PARTITION BY salesperson
            ORDER BY sale_date ASC) AS nextday_revenue
FROM sales;
```

sale_id	Salesperson	Sale_date	Revenue	nextday_revenue
1	Mbali	2026-08-01	1000	1200
2	Mbali	2026-08-02	1200	900
3	Mbali	2026-08-03	900	NULL
4	John	2026-08-04	800	950
5	John	2026-08-05	950	NULL

sale_id	Salesperson	Sale_date	Revenue
1	Mbali	2026-08-01	1000
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3	Mbali	2026-08-03	900
4	John	2026-08-04	800
5	John	2026-08-05	950

```

SELECT salesperson,
       sale_date,
       revenue,
       LAG (revenue)
         OVER (PARTITION BY salesperson
              ORDER BY sale_date ASC) AS prevday_revenue
FROM sales;

```

sale_id	Salesperson	Sale_date	Revenue	prevday_revenue
1	Mbali	2026-08-01	1000	NULL
2	Mbali	2026-08-02	1200	1000
3	Mbali	2026-08-03	900	1200
4	John	2026-08-04	800	NULL
5	John	2026-08-05	950	800